

SAMRAJ VISWASAM

DevOps Engineer | Cloud & Site Reliability

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PROFESSIONAL SUMMARY

DevOps Engineer with hands-on production experience building and operating CI/CD pipelines, Linux systems, containerized deployments, and Kubernetes environments on AWS (EKS, ECR, ALB), developed across a live internship and 4 end-to-end automation projects. Proficient in Python and Java, with a Computer Science foundation in object-oriented design, operating systems, algorithms, and complexity analysis. Skilled in Jenkins, GitHub Actions, Docker, Kubernetes, Git, AWS, and Azure fundamentals, with hands-on monitoring and alerting experience using Prometheus, Grafana, and Alertmanager. Reduced manual release steps, cut build and deployment time, and delivered reusable Infrastructure-as-Code pipelines. SUSE-certified in Kubernetes Administration (KUB201) and Microsoft-certified in Azure Data Fundamentals (DP-900). Seeking a DevOps / SRE / Cloud Engineer role in a large-scale software development environment.

TECHNICAL SKILLS

Operating Systems: Linux (Ubuntu, CentOS/RHEL), Unix, Shell/Bash Scripting, System Administration, Process & Service Management

Programming: Python, Java, JavaScript, SQL, YAML, Groovy (Jenkins Pipeline)

CI/CD & Automation: Jenkins, Jenkinsfile (Declarative Pipeline), GitHub Actions, GitHub Webhooks, Self-Hosted Runners, Maven, npm, Build/Release/Infra Automation

Containers & Orchestration: Docker, Dockerfile, Docker Compose, Container Registries, Image Optimization, Kubernetes (Deployments, Services, Secrets, PVCs, Ingress, HPA, ClusterIP, Minikube) - SUSE KUB201 Certified

Cloud: AWS (EC2, S3, IAM, EKS, ECR, Application Load Balancer/ALB Ingress), Microsoft Azure (Data Fundamentals - DP-900), Cloud Deployment, Virtualization, Cloud Computing

Monitoring & Observability: Prometheus, Grafana, Alertmanager, Dashboards, Logging, Alerting, Troubleshooting, Root Cause Analysis

Version Control: Git, GitHub, Branching Strategies, Pull Requests, Code Review

Databases: MySQL, PostgreSQL, RDBMS, DBMS, SQL Queries

CS Fundamentals: OOD, Data Structures, Algorithms, Complexity Analysis, Operating Systems, Design Patterns, Computer Networks, Distributed Systems, Reliability & Scaling

Practices: CI/CD, Infrastructure as Code, Configuration Management, Agile/Scrum, Documentation, On-Call Troubleshooting, Incident Response, SLA/SLO Awareness

PROFESSIONAL EXPERIENCE

DevOps Engineer Intern | Icanio Technologies, India

June 2026 - Present

- Built and maintained 4+ CI/CD pipelines using Jenkins and GitHub Actions, automating source-code checkout, dependency installation, testing, image builds, and container deployment - reducing manual release steps from ~8 to 1 per deployment.
- Administered 5+ Linux servers (user/permission management, package installation, service configuration, disk and log analysis) and wrote 10+ Bash scripts to automate repetitive operations tasks, cutting routine admin time by an estimated 30%.
- Containerized 3+ applications with Docker by authoring optimized Dockerfiles, reducing image size by ~35% and standardizing image builds, tagging, and registry pushes across dev, test, and production environments.
- Configured Prometheus and Grafana dashboards tracking 10+ system and application metrics, cutting average issue detection and triage time by an estimated 40% - supporting on-call incident response and root cause analysis.
- Analyzed application and system logs to troubleshoot build failures, container runtime errors, and deployment defects, documenting 6+ fixes as reusable runbooks for the team.
- Collaborated with developers to standardize pipeline-as-code in Git, reducing new-pipeline setup time and cutting redundant scripting across 3+ projects.

AI & ML Intern | Kodacy

July 2024 - August 2024

- Developed machine learning models in Python across the full workflow - data preprocessing, feature engineering, model training, and evaluation - on a dataset of 1,000+ records.
- Applied data structures, algorithms, and complexity analysis to optimize preprocessing routines, improving model testing throughput by an estimated 20%.

TECHNICAL PROJECTS

End-to-End CI/CD Deployment of a React and Flask Full-Stack Application on AWS EKS

Jenkins, GitHub, Docker, Amazon ECR, Amazon EKS, Kubernetes, AWS ALB Ingress

- Designed and automated an end-to-end CI/CD pipeline using Jenkins, GitHub, and Docker to build, version, and deploy React and Flask container images to Amazon ECR and Amazon EKS.
- Implemented Kubernetes Deployments, multi-replica workloads, and ClusterIP Services with automated rollout verification using kubectl for reliable, repeatable application delivery.
- Configured AWS Application Load Balancer (ALB) Ingress with path-based routing, exposing the React frontend at / and the Flask API at /api through a single public ALB.
- Delivered a cloud-native deployment pattern that eliminated manual image tagging and kubectl apply steps, standardizing releases to a single pipeline trigger.

End-to-End Automated CI/CD Pipeline for a Dockerized Application

Jenkins, Jenkinsfile, GitHub Webhooks, Docker, Linux, ngrok

- Designed a fully automated Jenkins Declarative Pipeline (pipeline-as-code via Jenkinsfile) triggered by GitHub Webhooks, enabling continuous integration on every commit and eliminating 100% of manual build initiation.
- Automated source-code checkout, dependency installation, testing, Docker image creation, and container deployment across 5 discrete, reusable pipeline stages with independent failure handling.
- Integrated Jenkins with the Docker engine and exposed the local server through ngrok, establishing reliable webhook connectivity that reduced build-trigger latency to under 10 seconds.
- Structured the pipeline with modular design patterns, making stages reusable across 3+ environments (dev, test, deployment).

CI/CD Deployment of a React Application using GitHub Actions and Docker

React.js, GitHub Actions, Docker, Linux, Self-Hosted Runner

- Architected a GitHub Actions CI/CD workflow with a self-hosted Linux runner, automating build, image creation, and deployment for a containerized React.js application.
- Tuned Docker build and runtime configuration, reducing image size by ~30% and shortening build times by an estimated 25%.
- Replaced a 6-step manual build-and-deploy process with a version-controlled YAML workflow, cutting deployment effort to a single commit.
- Configured runner environment, secrets handling, and job dependencies to keep the deployment process secure and repeatable across releases.

EDUCATION

B.Tech, Computer Science and Engineering

August 2022 - May 2026

Karunya Institute of Technology and Sciences, Coimbatore, India

Relevant Coursework: Data Structures and Algorithms, Design and Analysis of Algorithms (Complexity Analysis), Operating Systems, Object-Oriented Programming, Computer Networks, Database Management Systems, Software Engineering, Cloud Computing.

CERTIFICATIONS

- KUB201v1: Kubernetes Administration - SUSE
- DP-900: Microsoft Azure Data Fundamentals - Microsoft (Score: 790/1000, Oct 2024)
- Google AI Essentials - Google / Coursera
- Python Programming - Cisco Networking Academy
- JavaScript Essentials 1 & 2 - Cisco Networking Academy
- Java Essentials - Infosys Springboard
- Introduction to Industry 4.0 and Industrial Internet of Things - NPTEL

LANGUAGES

English (Professional), Tamil (Native)